

1 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/idealized_sbl/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5p`, `ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}`, `Nothing`, `typeof(OrdinaryDiffEqCore.DEFAULT_PRECS)`, `Val{:forward}()`, `true`, `nothing`, `typeof(OrdinaryDiffEqCore.trivial_limiter!)`, `typeof(OrdinaryDiffEqCore.trivial_limiter!)` and returned `Success`.

1.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/idealized_sbl`
- Number of saved time samples: 1081
- Number of profile snapshots: 4
- Solver accepted/rejected steps: 438 / 2
- RHS evaluations: 109956
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reltol} = 1.000 \times 10^{-6}$

1.2 Forcing and Boundary Conditions

Table 1: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$10.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	n/a m
Surface roughness (heat)	z_{0h}	n/a m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	$300.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\text{min,surf}}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

1.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.184562, 108.426452] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.007283, 43.999865]$

1.4 Generated Artifacts

- Time-series CSV: `results/idealized_sbl/time_series.csv`
- Diagnostic summary JSON: `results/idealized_sbl/summary.json`
- Payload JLD2: `results/idealized_sbl/payload.jld2`

1.5 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

1.6 Included Plots

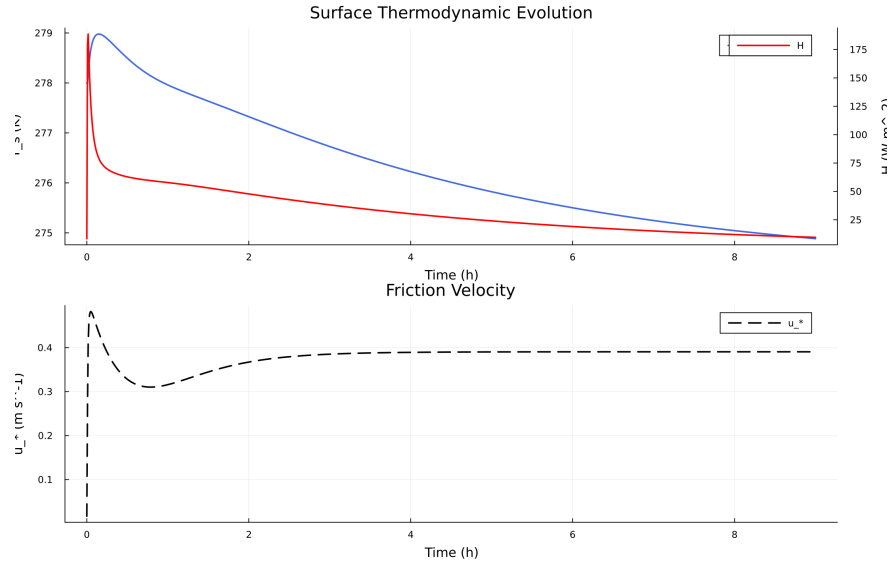


Figure 1: Time series of T_s , sensible heat flux, and friction velocity

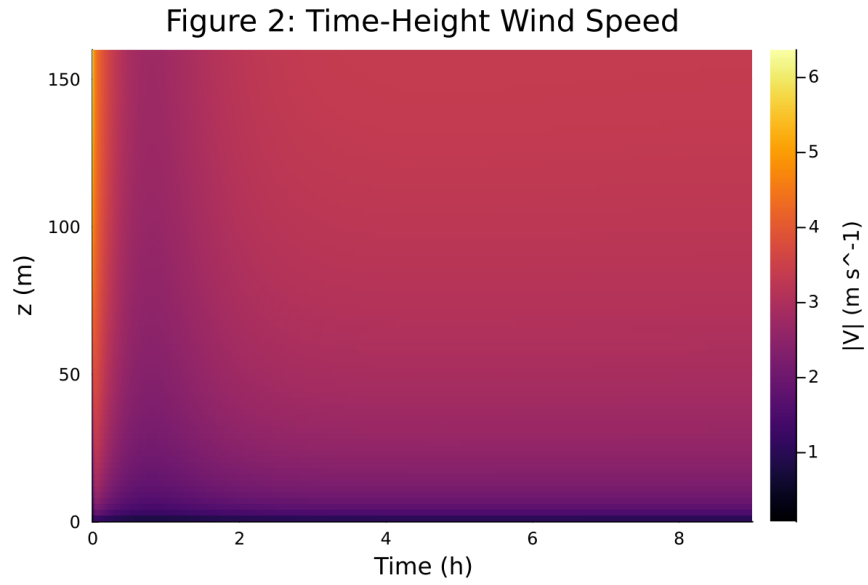


Figure 2: Time-height contour of wind speed with LLJ evolution

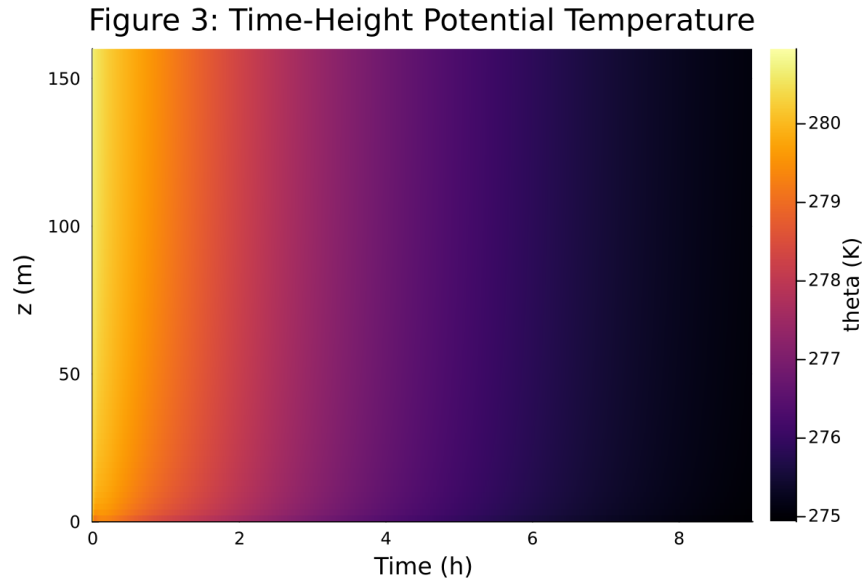


Figure 3: Time-height contour of potential temperature and inversion growth

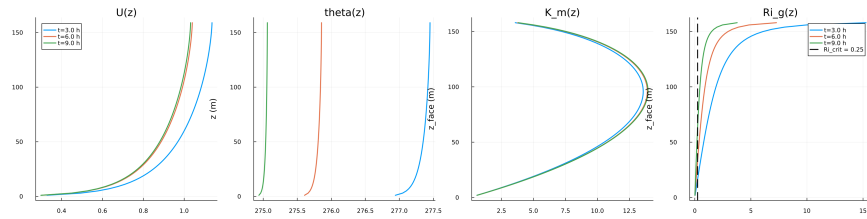


Figure 4: Vertical profiles of U , θ , and K_m at representative times

Figure 5: Surface Energy Budget

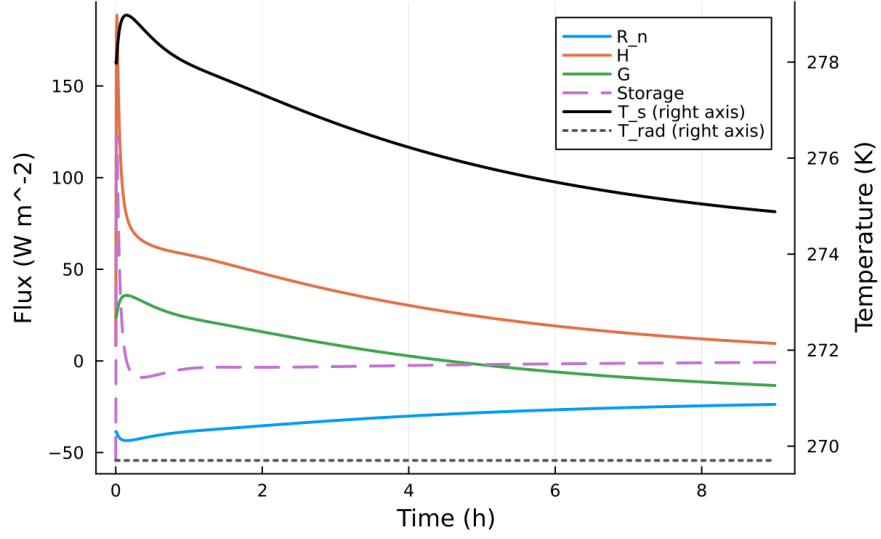


Figure 5: Surface energy budget components (R_n , H , G , storage)

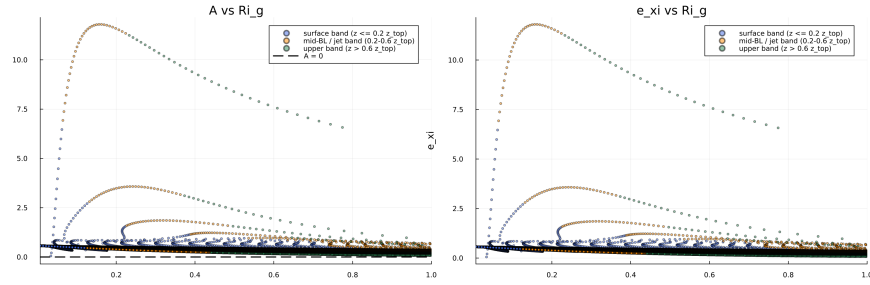


Figure 6: Phase portrait on regularized manifold (Δ vs e_{xi})

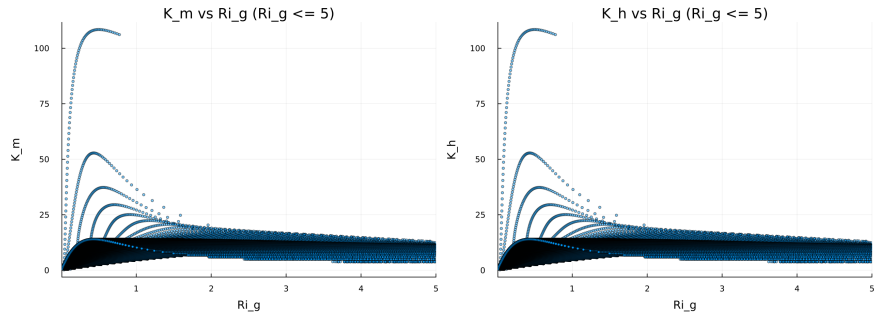


Figure 7: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

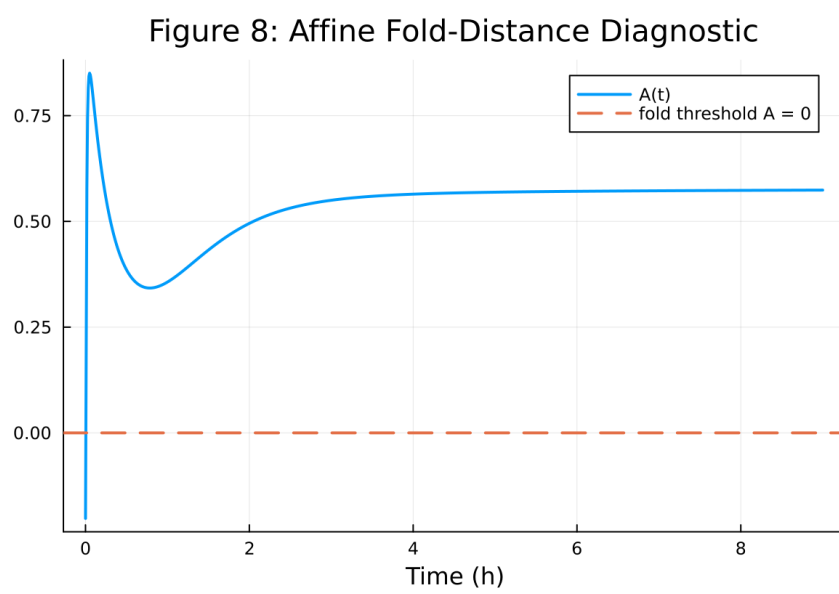


Figure 8: Fold proximity diagnostics versus time